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LEARNER GUIDE

For

Sustainability Masterclass

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v1.0	1 October 2026	Initial release of the Sustainability Masterclass learner guide: standards, Singapore context and nine real-use-case activities	Dr Alfred Ang

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How to Use This Learner Guide

This guide contains the detailed concepts, activity procedures, scenarios and questions. The slide deck is deliberately visual and concise; use this guide for procedural detail, calculations and evidence expectations.

Course Information

Course: Sustainability Masterclass

Course code: C752

Duration: 15 instructional hours over 2 days (9:30am – 5:30pm)

Trainer: Dr Alfred Ang

Learning Outcomes

- LO1 — Analyse carbon emission hotspots using recognised standards, boundaries and reliable activity data.
- LO2 — Prioritise carbon-footprint reduction initiatives using projections, feasibility, cost-benefit and decision tools.
- LO3 — Develop an implementable carbon-footprint reduction plan with governance, targets, monitoring and continual improvement.

Topic 1: Analyse Carbon Emission Hotspots

Standards, boundaries, data, scopes and hotspot diagnosis

Learning outcome: LO1

A Carbon Inventory Is a Decision System

Purpose: Translate physical activity into comparable tCO_2e for decisions.

Boundary: State which entities, operations, gases and period are included.

Method: Use consistent activity data, emission factors and global-warming potentials.

Evidence: Retain sources, assumptions, controls and recalculation rules.

Build a Corporate Inventory in Five Moves

1. Set organisational boundary
2. Set operational boundary
3. Collect activity data
4. Calculate and consolidate
5. Report, verify and improve

Five Principles Protect Decision Quality

Relevance: Reflect the organisation and users' needs.

Completeness: Account for all sources within the boundary; disclose exclusions.

Consistency: Enable meaningful comparison over time.

Transparency: Disclose methods, assumptions and evidence.

Accuracy: Reduce bias and uncertainty as far as practical.

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Organisational Boundary Choices

Equity share	Control approach
Account according to economic interest	Account for operations under financial or operational control
Useful where ownership shares matter	Often aligns with management systems
Apply consistently across investments	Document the chosen control test

How the Standards Fit Together

1. ISO 14001 structures the EMS
2. GHG Protocol structures corporate accounting
3. ISO 14064-1 supports organisation-level quantification
4. ISO 14064-3 supports verification
5. Assurance strengthens confidence

Accounting Standard vs Management Standard

GHG Protocol / ISO 14064	ISO 14001 EMS
What and how much is emitted	How environmental performance is managed
Boundaries, categories, calculations	Policy, objectives, controls and review
Inventory quality and reporting	Continual improvement through PDCA

The Seven Greenhouse Gases

CO_2: Fuel and process emissions

CH_4: Waste, agriculture and fugitive emissions

N_2O: Combustion, fertiliser and processes

HFCs: Refrigeration and cooling

PFCs: Electronics and industrial processes

SF_6: Electrical equipment

NF_3: Electronics and semiconductor manufacturing

From Activity to tCO_2e

A transparent inventory keeps every assumption visible.

Emissions (tCO_2e) = Activity data × Emission factor × GWP ÷ unit conversion

- Example: kWh × kg CO_2e/kWh ÷ 1,000
- Use the factor for the correct geography and year
- Record factor source, unit and uncertainty

A Practical Data-Quality Ladder

1 · Supplier-specific: Verified primary data for the exact product or service.

2 · Metered activity: Invoices, meters, fuel records or logistics data.

3 · Engineering estimate: Measured drivers with documented assumptions.

4 · Spend proxy: Currency × environmentally extended input-output factor.

5 · Unsupported guess: Use only temporarily; flag for improvement.

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Three Scopes Prevent Category Confusion

Scope 1: Direct emissions from sources owned or controlled.

Scope 2: Indirect emissions from purchased electricity, steam, heat or cooling.

Scope 3: Other value-chain emissions upstream and downstream.

Rule: Scopes describe responsibility for accounting—not moral blame.

Location-Based and Market-Based Views

Location-based	Market-based
Average grid emissions where consumption occurs	Contractual instruments and supplier attributes
Useful for physical system context	Requires quality criteria and evidence
Singapore example: EMA grid factor	Report both where applicable

Follow the Value Chain, Not the Org Chart

Upstream: Purchased goods, capital goods, fuels, transport, waste, travel, commuting and leased assets.

Downstream: Distribution, processing, use, end-of-life, leased assets, franchises and investments.

Screen first: Estimate all relevant categories before investing in precision.

Engage next: Improve supplier and customer data at material hotspots.

The Hotspot Analysis Funnel

1. Create source register
2. Estimate complete footprint
3. Rank absolute tCO_2e
4. Test intensity and trend
5. Investigate controllability and risk

Illustrative Footprint by Source

Category	Illustrative value
Materials	44
Electricity	26
Freight	15
Fuel	10
Waste	5

Top two sources represent 70% of the screening footprint; validate them first.

A Hotspot Is More Than the Largest Bar

Magnitude: Absolute or intensity-based share of emissions.

Influence: Ability to change technology, demand or supplier behaviour.

Risk: Carbon price, regulation, disruption or reputation.

Strategic fit: Connection to customers, growth and operational priorities.

Common Inventory Failure Modes

Missing sources: Completeness check against sites, ledgers and processes.

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Double counting: Clear ownership, category definitions and consolidation rules.

Unit errors: Automated validation and unit conversion controls.

Wrong factor: Versioned factor library with geography and year.

Silent estimates: Estimate register with owner and replacement date.

Can the SG Carbon Calculator Be Used Here?

Is the objective personal engagement or a corporate inventory?

If yes / fit: Personal engagement: use calculator outputs to start conversations and identify behaviours.

If no / not yet: Corporate inventory: use controlled organisational activity data, documented factors and boundary rules.

The calculator is a screening and learning aid—not a substitute for a GHG Protocol corporate inventory.

Reading Singapore Electricity Data Correctly

Published factor: EMA reported 0.402 kg CO_2/kWh for 2024.

Correct year: Match the factor year to the reporting period.

Correct unit: Convert kg to tonnes only once.

Record evidence: Save the factor source and access date.

Recalculate: Define when base-year electricity emissions are restated.

Separate Carbon Value From Cash Value

Carbon effect: Annual and lifetime tCO_2e avoided.

Cash effect: Capex, opex, savings and residual value.

Risk effect: Carbon tax, energy volatility and compliance exposure.

Strategic effect: Resilience, customer value and learning.

Simple Payback Is Fast—but Incomplete

Useful as a screening metric when cash flows are stable.

Simple payback (years) = Initial investment ÷ Annual net cash saving

- Ignores time value of money
- Ignores cash flows after payback
- Never rank long-lived projects by payback alone

NPV Makes Timing Explicit

Discount future net cash flows to present value.

$NPV = \Sigma\ [\text{Net cash flow}_t \div (1+r)^t] - \text{Initial investment}$

- Positive NPV creates financial value at the chosen discount rate
- Test energy price, utilisation and maintenance assumptions
- Use sensitivity analysis instead of false precision

Compare Projects per Tonne Avoided

Marginal abatement cost exposes the price of each carbon outcome.

MAC = Lifetime net cost ÷ Lifetime tCO_2e avoided

- Negative cost means the measure saves money
- Use consistent lifetime and discounting
- Do not ignore feasibility and dependencies

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Illustrative Abatement Portfolio

Category	Illustrative value
LED	-55
Controls	-20
Solar	18
EV fleet	65
Green material	110

Negative-cost measures can help fund strategic measures with positive abatement cost.

Stress-Test the Variables That Matter

Energy price: Low, expected and high tariff cases.

Utilisation: Operating hours, throughput and occupancy.

Performance: Degradation, rebound and implementation shortfall.

Carbon value: Tax, internal carbon price or customer premium.

Discount rate: Financing and risk assumptions.

Two Ways to Price Carbon Internally

Shadow price	Internal fee
Apply a hypothetical carbon value in investment cases	Charge emissions and recycle funds into reductions
Changes decisions without charging business units	Creates a visible operating signal
Good for stress-testing long-lived assets	Needs governance and acceptance

LED Retrofit Decision

Illustrative teaching data: S$120,000 capex; S$38,000 annual saving; 72 tCO_2e avoided per year; 8-year life.

Payback = 120,000 ÷ 38,000 = 3.16 years
Simple lifetime MAC = (120,000 − 8×38,000) ÷ (8×72) = −S$319/$tCO_2e$

- The project saves money and carbon
- Confirm maintenance, hours and baseline
- Use discounted cash flow for approval

From Standards to a Defensible Hotspot

Frame: Purpose, organisational boundary, operational boundary and period.

Quantify: Activity data × factor × GWP with controlled units.

Diagnose: Rank magnitude, trend, risk and controllability.

Value: Combine tonnes, cash, risk and strategic fit.

Evidence Should Survive the Handover

1. Source register
2. Calculation workbook
3. Quality checks
4. Hotspot narrative

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5. Investment assumptions

Precision without completeness can produce a confidently wrong hotspot.

Precision without completeness can produce a confidently wrong hotspot.

Screen broadly, then improve the data where decisions are material.

Topic 1 Activities

Activity 1: Map a Multi-Site Corporate Inventory

Real-use-case lens	Singapore retail and distribution group
Source lens	GHG Protocol Corporate Standard
Objective	Define organisational and operational boundaries, classify sources and identify data gaps.
Deliverable	Boundary memo, scope map and data-gap register

Scenario

Merlion Retail Group operates 12 stores, one warehouse, a leased head office and a contracted delivery fleet. It owns 80% of a Malaysian sourcing subsidiary. Management wants a 2025 baseline before setting a target.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Read the scenario and identify the intended users of the inventory.
2. Draw the legal and operating structure, including leased and contracted activities.
3. Choose equity share, financial control or operational control; state the rationale.
4. Set the reporting period, gases and materiality rule.
5. Create a source register for stores, warehouse, office, subsidiary and logistics.
6. Classify sources by scope and relevant Scope 3 category.
7. Rate each source on magnitude, evidence quality and controllability.
8. Record exclusions, assumptions and the conditions that would trigger recalculation.
9. Peer-review the boundary memo against the five GHG Protocol principles.
10. Present the two highest-risk data gaps and an improvement action for each.

Scenario questions

1. Which consolidation approach should the group adopt, and why?
2. Classify each source into Scope 1, 2 or 3.
3. Which exclusions would be material and require disclosure?
4. What evidence is needed to replace the three weakest estimates?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.

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- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 2: Use the SG Carbon Calculator as an Engagement Tool

Real-use-case lens	Employee carbon-literacy campaign
Source lens	SG Carbon Calculator and UN ActNow
Objective	Use personal-footprint screening responsibly, then translate behavioural insights into corporate data needs.
Deliverable	Calculator evidence, limitations note and engagement experiment

Scenario

A 450-person professional-services firm wants an employee campaign covering commuting, home energy, food and waste. The sustainability lead proposes using the SG Carbon Calculator and adding all results to the corporate inventory.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Open https://alfredang.github.io/sgcarboncalculator/ in a modern browser.
2. Record a fictional persona; do not enter identifiable learner data.
3. Complete the energy, transport, food and waste inputs.
4. Capture the total and category breakdown in the provided worksheet.
5. Change one material input and observe the direction and scale of change.
6. Select two relevant UN ActNow actions and explain the mechanism of impact.
7. Map each action to corporate Scope 1, 2 or 3 influence.
8. List the calculator's boundary, factor and verification limitations.
9. Design a four-week opt-in engagement experiment with one aggregate outcome metric.
10. Write a clear disclaimer separating personal screening from corporate accounting.

Scenario questions

1. What can the calculator reliably support?
2. Why should personal results not be summed into a corporate inventory?
3. Which actions connect to Scope 3 categories?
4. How will you measure participation without collecting unnecessary personal data?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 3: Find Hotspots in a Farm-to-Table Value Chain

Real-use-case lens	CP Foods-inspired teaching case

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Source lens	SAP customer spotlight and GHG Protocol
Objective	Calculate a screening footprint, rank hotspots and propose a data-improvement plan.
Deliverable	Calculated footprint, Pareto chart and hotspot recommendation

Scenario

Orchid Foods produces chilled ready meals for Singapore retailers. Inspired by CP Foods' farm-to-table footprint work, it wants a first screening of ingredients, factory energy, cold-chain logistics, packaging and waste. All figures are illustrative teaching data, not reported CP Foods results.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Review the activity-data table and check every unit.
2. Match each row to the supplied illustrative emission factor.
3. Calculate tCO_2e for every source and retain the formula trail.
4. Classify each result by scope and value-chain stage.
5. Calculate total footprint and percentage contribution.
6. Rank sources and draw a Pareto chart.
7. Normalise the result per 1,000 meals and per S$1 million revenue.
8. Rate the top five sources by data quality and controllability.
9. Select two priority data improvements and assign owners.
10. Present a hotspot narrative that distinguishes calculation result from management judgement.

Scenario questions

1. Which two sources dominate total tCO_2e?
2. How would results change under a different functional unit?
3. Where is supplier-specific data most valuable?
4. Which hotspot is largest but least controllable?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

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Topic 2: Prioritise Reduction Initiatives

Projection, technology, economics and transparent decisions

Learning outcome: LO2

A Baseline Is a Managed Reference Point

Base year: Representative, reliable and relevant to the target.

Structural change: Acquisition, divestment or outsourcing may require recalculation.

Method change: Material factor or methodology changes may require restatement.

Growth: Separate absolute emissions from intensity and activity growth.

Start With the Business-as-Usual Case

Make growth, efficiency and factor assumptions explicit.

Future emissions = Future activity × Future emission factor × (1 − efficiency improvement)

- Build the no-action case first
- Avoid claiming reductions caused by grid decarbonisation as operational savings
- Use compatible units and periods

Absolute and Intensity Targets Answer Different Questions

Absolute target	Intensity target
Total tCO_2e falls against baseline	tCO_2e per unit of output or revenue
Directly aligned to atmospheric outcome	Separates growth from efficiency
Can challenge a fast-growing business	Can improve while total emissions rise

Build Scenarios as Decision Stories

1. Define drivers
2. Choose coherent assumptions
3. Model business-as-usual
4. Layer reduction measures
5. Compare emissions, cost and risk

Use Ranges Where the Future Is Uncertain

Low case: Slower growth, lower adoption or conservative performance.

Expected case: Best current estimate with documented evidence.

High case: Faster growth, price shock or implementation upside.

Trigger: Define the observation that causes a plan revision.

Illustrative 2030 Emissions Path

Category	Illustrative value
2025	100
2026	98
2027	91
2028	83
2029	74
2030	65

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A target path becomes actionable only when initiatives, owners and milestones explain each step down.

Explain Why Emissions Changed

Separate operational performance from structural and market effects.

Δ Emissions = Activity effect + Efficiency effect + Factor effect + Structural effect

- Activity effect captures volume or growth
- Efficiency effect captures operational improvement
- Factor effect captures energy or supplier decarbonisation

Avoid the Rebound Trap

Efficiency gain: Less energy per service unit.

Behaviour response: Lower cost can increase demand or operating hours.

Net result: Gross saving minus additional consumption.

Control: Track both intensity and absolute activity.

Electrifying a Singapore Delivery Fleet

Illustrative: 20 vans × 35,000 km/year; diesel van 0.25 kg CO_2e/km; EV 0.20 kWh/km; grid factor 0.402 kg CO_2/kWh.

Diesel = 175 tCO_2e/year
EV electricity = 56.3 tCO_2e/year
Screening reduction = 118.7 tCO_2e/year

- Add vehicle and battery lifecycle where material
- Model future grid factors separately
- Validate actual route, charging and payload data

Reduce Before You Compensate

Avoid: Remove unnecessary demand, travel, material or process.

Reduce: Improve efficiency and yield.

Substitute: Use lower-carbon energy, material or transport.

Engage: Work with suppliers, customers and employees.

Neutralise residuals: Use credible removals only after deep reductions.

Energy Levers: Measure, Control, Replace

Measure: Sub-meter material loads and normalise for production.

Control: Schedules, set-points, maintenance and automation.

Upgrade: Efficient motors, chillers, compressed air and heat recovery.

Replace: Renewable electricity and electrified heat where feasible.

Operational Excellence Is Carbon Management

Yield: Prevent scrap and rework.

Flow: Reduce waiting, idle loads and unnecessary movement.

Maintenance: Keep assets at designed efficiency.

Planning: Match batches, routes and capacity to demand.

Design: Remove carbon before it is locked into the process.

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Supplier Engagement Needs a Data Ladder

Screen: Use secondary factors to find material categories.

Request: Ask material suppliers for product or process data.

Validate: Check boundary, method, period and assurance.

Collaborate: Co-design lower-carbon specifications and logistics.

Contract: Embed evidence and performance expectations.

Circular Design Changes the Carbon Equation

1. Refuse unnecessary input
2. Reduce material intensity
3. Reuse components
4. Repair and remanufacture
5. Recycle at highest practical value

Industry 4.0 Can Help—and Create New Risks

Opportunity	Implementation risk
IoT enables real-time energy and process data	High integration cost and interoperability gaps
Analytics detects anomalies and predicts failures	Data security and ownership concerns
ERP connects carbon to materials, orders and suppliers	Automation can scale poor-quality data

What an Enterprise Footprint Platform Connects

Transactions: Materials, purchase orders, production and logistics.

Factors: Versioned libraries matched to geography and period.

Flows: Energy and material movement across products and sites.

Results: Corporate and product footprints with drill-down.

Action: Hotspots integrated into operational decisions.

Is the Technology Ready for This Site?

Does the measure meet technical, operational, financial and evidence thresholds?

If yes / fit: Pilot with success criteria, monitoring plan and scale-up gate.

If no / not yet: Redesign, sequence a dependency, or retain in the opportunity backlog.

A promising technology is not yet a feasible initiative until site constraints and implementation risk are tested.

Design a Pilot That Can Be Judged

Baseline: Comparable pre-intervention period and operating conditions.

Intervention: Specific asset, control or behaviour change.

Metric: Energy, output, emissions, cost and quality guardrail.

Duration: Long enough to capture representative variation.

Decision rule: Scale, modify or stop threshold agreed in advance.

A Portfolio Needs More Than One Score

Impact: Annual and cumulative tCO_2e.

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Economics: NPV, payback and abatement cost.

Feasibility: Technology maturity, resources and disruption.

Strategic fit: Customer, compliance and resilience value.

Confidence: Data quality and dependency risk.

A Transparent Multi-Criteria Decision Process

1. Agree criteria
2. Set weights
3. Define scoring anchors
4. Score with evidence
5. Test sensitivity and approve

Make Trade-Offs Visible

Use a common scale and documented evidence.

Weighted score = Σ (criterion score × criterion weight)

- Weights must total 100%
- Use behavioural anchors for 1–5 scores
- Do not let the formula replace governance

No-Regret Measures and Strategic Bets

No-regret	Strategic bet
Strong economics and proven technology	Higher uncertainty or capital need
Low dependency and quick implementation	Enables deep reductions or future compliance
Build momentum and fund later work	Use staged gates and learning milestones

What Should Start First?

Is the measure high-impact, feasible now and independent of another project?

If yes / fit: Place in the immediate delivery wave with a named owner and budget gate.

If no / not yet: Place in enable, pilot, partner or monitor wave with a trigger date.

Sequence dependencies: data platform before optimisation, charging before fleet conversion, specification before supplier tender.

Illustrative Initiative Scores

Category	Illustrative value
Controls	88
Solar	76
Fleet EV	69
Supplier data	64
Green steel	52

A score supports discussion; the approval record must still explain constraints, dependencies and dissent.

Counter Common Decision Biases

Payback bias: Include lifecycle value and residual risk.

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Pet-project bias: Require comparable evidence and scoring anchors.

Precision bias: Show ranges and confidence ratings.

Silo bias: Include operations, finance, procurement and users.

Offset bias: Apply the reduction hierarchy first.

Portfolio Choice Under a S$500k Cap

Illustrative measures: controls S$80k/160 t; solar S$280k/210 t; EV pilot S$140k/90 t; green material S$220k/300 t.

Possible Wave 1: controls + solar + EV pilot = S$500k and 460 tCO_2e/year screened benefit

- Green material has high impact but needs procurement validation
- Check double counting and dependencies
- Retain a reserve for measurement and commissioning

Prioritisation Is a Portfolio Conversation

Project: Model baseline, intervention and uncertainty.

Prove: Test technical and operating feasibility.

Compare: Use carbon, cash, risk, strategy and confidence.

Sequence: Balance no-regret delivery with deep-reduction enablers.

From Long List to Funded Roadmap

1. Opportunity register
2. Business cases
3. MCDA scoring
4. Dependency map
5. Approved waves

The best portfolio is credible enough to fund and flexible enough to learn.

The best portfolio is credible enough to fund and flexible enough to learn.

Use scenarios, pilots and stage gates to turn uncertainty into managed evidence.

Topic 2 Activities

Activity 4: Model a Singapore Fleet Transition

Real-use-case lens	LionCity Logistics
Source lens	EMA grid statistics and GHG Protocol
Objective	Build business-as-usual and reduction scenarios for diesel, route optimisation and fleet electrification.
Deliverable	2030 scenario model, assumptions register and recommendation

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Scenario

LionCity Logistics runs 40 diesel vans in Singapore. Customer demand is expected to grow 5% per year. Management is considering route optimisation, driver coaching and replacement of 20 vans with EVs by 2028.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Review the fleet, distance, fuel and growth assumptions.
2. Calculate the 2025 diesel baseline in tCO_2e.
3. Project annual vehicle-kilometres to 2030 under business-as-usual.
4. Model route optimisation as an activity reduction.
5. Model driver coaching as an efficiency improvement.
6. Model EV electricity using 0.20 kWh/km and the stated grid-factor scenarios.
7. Create low, expected and high cases for adoption and performance.
8. Separate activity, efficiency, factor and structural effects.
9. Run sensitivity on annual distance and grid factor.
10. Recommend a phased roadmap with one decision trigger per phase.

Scenario questions

1. What is the business-as-usual 2030 footprint?
2. Which reduction is operational and which depends on the electricity factor?
3. What rebound or capacity risks could erase savings?
4. What trigger would change the fleet plan?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 5: Screen a Smart-Factory Carbon Initiative

Real-use-case lens	Matsumoto Precision-inspired teaching case
Source lens	SAP customer spotlight and Yurtay (2025)
Objective	Assess the technical, operational, data and financial feasibility of IoT/ERP-enabled energy optimisation.
Deliverable	Feasibility scorecard and pilot charter

Scenario

Sakura Precision SG operates CNC machining, compressed air and heat-treatment equipment. Inspired by Matsumoto Precision's public sustainability-footprint work, it is considering sub-metering, IoT sensors and ERP integration. All figures are illustrative.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

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Detailed procedure

1. Map the energy and material flows for the selected production cell.
2. Identify decisions currently blocked by missing or delayed data.
3. Define the minimum sensors, tags and ERP master data needed.
4. Assess interoperability, cybersecurity, ownership and maintenance risks.
5. Estimate baseline energy, idle load and scrap-related emissions.
6. Define the intervention and control group or comparison period.
7. Calculate screened savings, cost and payback range.
8. Set quality, uptime and production guardrails.
9. Score technical, operational, financial and evidence feasibility.
10. Write a 12-week pilot charter with scale, modify and stop rules.

Scenario questions

1. Which data should be integrated first?
2. What is the minimum viable pilot?
3. Which risks could scale a bad result?
4. What success criteria justify rollout?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 6: Prioritise a Reduction Portfolio

Real-use-case lens	WITTE Automotive-inspired teaching case
Source lens	SAP customer spotlight, CarbonChain and IMD
Objective	Combine MACC, weighted criteria, dependencies and budget constraints into a transparent portfolio.
Deliverable	MACC, MCDA matrix, funded roadmap and decision record

Scenario

A Singapore automotive-components supplier has six shortlisted initiatives and S$700,000 available in Wave 1. The case is inspired by public examples of enterprise footprint management; all decision data are illustrative.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Check initiative boundaries and remove any overlapping savings.
2. Calculate lifetime tonnes avoided and lifetime net cost.
3. Calculate marginal abatement cost for each initiative.
4. Agree weights for carbon, NPV, feasibility, strategy and confidence.

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5. Define 1–5 scoring anchors before scoring.
6. Score each initiative using the evidence supplied.
7. Run a sensitivity case that doubles the carbon-impact weight.
8. Map technology, data, procurement and infrastructure dependencies.
9. Select a Wave 1 portfolio within the S$700,000 cap.
10. Document funded, pilot, enable and defer decisions with reasons.

Scenario questions

1. Which initiatives have negative abatement cost?
2. How sensitive is the ranking to weights?
3. Which dependency changes the sequence?
4. Which high-impact measure should be piloted rather than rejected?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

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Topic 3: Develop the Reduction Plan

EMS governance, action planning, Singapore requirements and review

Learning outcome: LO3

An EMS Turns Intent Into Repeatable Control

Context: Interested parties, risks, opportunities and obligations.

Leadership: Policy, accountability and resources.

Planning: Aspects, objectives, targets and actions.

Support & operation: Competence, communication and operational controls.

Evaluation & improvement: Monitor, audit, review and correct.

Carbon Management Through PDCA

1. Plan boundary, baseline, risks and targets
2. Do controls, projects and engagement
3. Check data, KPIs, audits and variance
4. Act on findings and reset priorities

Ownership and Accountability Are Different

Process owner	Carbon / GHG manager
Controls the activity and operational data	Maintains method, inventory and controls
Implements agreed actions	Coordinates reporting and assurance
Explains performance variance	Escalates quality and compliance issues

Build Controls Around the Data Journey

Capture: Authorised sources and complete populations.

Transform: Locked factors, units and calculation rules.

Review: Threshold checks and independent approval.

Change: Version control and recalculation log.

Retain: Evidence repository and retention period.

The Plan Needs People Who Can Operate It

Executives: Approve ambition, capital and risk appetite.

Finance: Validate business cases and reporting controls.

Operations: Own sources, projects and root causes.

Procurement: Engage suppliers and embed specifications.

Data / IT: Maintain architecture, access and quality.

A Useful Review Drives Decisions

1. Review target trajectory
2. Explain material variance
3. Assess project delivery
4. Consider risk and obligations
5. Decide resources and corrective action

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Audit the System, Not Only the Number

Boundary: Is it complete and consistently applied?

Evidence: Can samples be traced to source?

Calculation: Are factors, units and formulas controlled?

Action: Are corrective actions owned and closed?

Communication: Are claims supported and balanced?

Is the Variance a Data Error or Performance Gap?

Does corrected evidence bring the KPI back within tolerance?

If yes / fit: Fix the data/control issue, assess restatement and prevent recurrence.

If no / not yet: Investigate operational root cause and implement a performance corrective action.

Do not hide a missed target by changing the data; do not launch a project to fix a calculation error.

A footprint becomes manageable when it has owners, controls and review rhythms.

A footprint becomes manageable when it has owners, controls and review rhythms.

The inventory is an operating process—not an annual spreadsheet event.

A Credible Reduction Plan Has Eight Parts

1 · Boundary and baseline: What is covered and measured.

2 · Hotspots: Where emissions concentrate and why.

3 · Target: Outcome, year, scope and trajectory.

4 · Initiatives: Measures, savings and dependencies.

5 · Roadmap: Sequence, milestones and resources.

6 · Governance: Owners and decision rights.

7 · Measurement: KPIs, evidence and review.

8 · Risk & communication: Uncertainty, claims and escalation.

A Target Needs Scope and Substance

Weak target	Decision-ready target
Reduce emissions significantly	35% absolute Scope 1+2 reduction by 2030 from 2025
No base year or boundary	Boundary and recalculation policy stated
No trajectory or owner	Annual trajectory and accountable executive

Sequence the Reduction Plan

1. Stabilise data and controls
2. Deliver no-regret efficiency
3. Pilot strategic technology
4. Transform supply and product choices
5. Address residual emissions

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Connect Outcome, Driver and Delivery Metrics

Outcome: Total tCO_2e and tCO_2e per functional unit.

Driver: kWh/unit, fuel/km, kg material/unit, waste yield.

Delivery: Project milestone, installed capacity, supplier coverage.

Quality: Primary-data share, estimate share and control exceptions.

Make Failure Modes Visible Before Launch

Delivery: Delay, skills, vendor or permit.

Performance: Savings lower than modelled.

Commercial: Tariff, price, demand or finance.

Data: Missing evidence or incompatible systems.

Claims: Overstatement, double counting or unclear offsets.

Reduction and Offset Are Not Interchangeable

Operational reduction	Offset / credit
Changes emissions within the inventory or value chain	Finances a reduction or removal outside the inventory
Traceable to a measure and baseline	Requires quality and retirement evidence
Supports the reduction trajectory	Disclose separately from gross emissions

A Responsible Claim Shows the Full Picture

Boundary: Which entities, scopes and period?

Metric: Gross, net, absolute or intensity?

Method: Which standard, factors and recalculations?

Progress: Actual result versus trajectory.

Limitations: Estimates, uncertainty and exclusions.

Translate a 35% Target Into Annual Milestones

Illustrative 2025 baseline: 10,000 tCO_2e. 2030 target: 6,500 tCO_2e.

Required reduction = 3,500 tCO_2e
Linear planning guide = 700 tCO_2e/year average

- Actual roadmap need not be linear
- Link each step change to initiatives
- Track business-as-usual growth separately

Budget the Enablers, Not Only the Hardware

Measurement: Meters, factors, assurance and data engineering.

People: Project management, training and supplier engagement.

Change: Communications, workflow redesign and incentives.

Contingency: Performance shortfall and dependency risk.

Maintenance: Commissioning and sustained optimisation.

Every Action Needs a Complete Contract

What: Specific intervention and boundary.

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Why: Hotspot and expected outcome.

Who: Accountable owner and support roles.

When: Start, milestone, completion and review.

How much: Budget, resources and quantified benefit.

Evidence: KPI, source and verification method.

Make Objectives SMARTER

A target should guide action and adaptation.

Specific · Measurable · Achievable · Relevant · Time-bound · Evaluated · Reviewed

- Use an outcome metric and at least one driver metric
- Name the evidence source and review cadence
- Predefine escalation and revision conditions

Clarify Roles Before Delivery Starts

1. Accountable owner approves
2. Responsible team delivers
3. Consulted experts shape decisions
4. Informed stakeholders receive updates

Use a Single Initiative Register

Identity: Initiative ID, source, site and owner.

Value: Capex, savings, tCO_2e and confidence.

Delivery: Stage, milestone, dependency and issue.

Decision: Approval, gate and rationale.

Evidence: Baseline, M&V method and realised result.

Should the Initiative Scale?

Did the pilot meet carbon, cost, operational and data-quality criteria?

If yes / fit: Approve rollout with updated benefit case and implementation controls.

If no / not yet: Modify or stop; document learning and update the opportunity register.

Stage gates protect capital while preserving learning from uncertainty.

Measure Realised Savings Fairly

Baseline model: Representative pre-project performance.

Adjustment: Production, weather, occupancy or route mix.

Boundary: Avoid leakage and double counting.

Persistence: Check that savings continue after launch.

Approval: Independent review of material results.

Know the Current Singapore Thresholds

Reportable facility: Annual direct emissions at or above 2,000 tCO_2e.

Taxable facility: Annual direct emissions at or above 25,000 tCO_2e.

2026–2027 tax rate: S$45 per tCO_2e.

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Toward 2030: Government has indicated S$50–S$80 per tCO_2e by 2030.

Scope: Industrial facilities in specified sectors; verify current CPA requirements.

Reportable and Taxable Facilities

≥2,000 tCO_2e direct	≥25,000 tCO_2e direct
Register as reportable where applicable	Also subject to carbon tax
Submit annual Emissions Report by 30 June following year	Monitoring Plan and verified Emissions Report
Appoint required representatives and manager	Stronger controls and verification requirements

Screen the 2026 Carbon-Tax Exposure

Illustrative taxable emissions: 30,000 tCO_2e. 2026 rate: S$45/$tCO_2e$.

Gross screened liability = 30,000 × S$45 = S$1,350,000

- Use verified reportable emissions and current law for filing
- Account for eligible allowances only with competent advice
- Use the result to stress-test investments—not as the inventory boundary

A Plan Is a Managed System

Govern: Owners, controls, competence and management review.

Plan: Boundary, target, initiatives, roadmap and resources.

Deliver: Action contracts, stage gates and dependencies.

Learn: Measurement, audit, corrective action and transparent claims.

From Target to Continual Improvement

1. Approve target
2. Fund roadmap
3. Deliver actions
4. Verify outcomes
5. Review and adapt

The reduction plan is credible when someone can execute it, finance it and verify it.

The reduction plan is credible when someone can execute it, finance it and verify it.

Targets become performance only through governance and evidence.

Topic 3 Activities

Activity 7: Prepare a Singapore MR and Carbon-Tax Readiness Plan

Real-use-case lens	Jurong specialty-chemicals facility
Source lens	NEA Carbon Tax and MR requirements
Objective	Screen regulatory status, data controls, reporting obligations and carbon-price exposure.
Deliverable	Obligations matrix, control plan and screened tax exposure

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Scenario

HarbourChem's Jurong facility estimates 30,000 tCO_2e of annual direct emissions from natural gas, process emissions and refrigerants. Management needs a readiness review for the 2026 reporting year. This is an illustrative learning case, not legal advice.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Read the current NEA carbon-tax and MR pages supplied in the Sources file.
2. Confirm the reporting year and the current threshold and rate used in the case.
3. Create a complete direct-emission source register.
4. Map each source to activity data, factor, method and evidence owner.
5. Screen annual tCO_2e and compare it with 2,000 and 25,000 tCO_2e thresholds.
6. Calculate gross carbon-tax exposure at the stated rate.
7. Draft a RACI for Designated Representative, GHG Manager, operations, finance and verifier.
8. Design capture, calculation, review, change-control and retention controls.
9. Build a calendar for monitoring-plan, verification and annual-report readiness.
10. Record legal and technical questions requiring current professional confirmation.

Scenario questions

1. Why is the facility both reportable and taxable?
2. Which sources belong in direct emissions?
3. What evidence and roles are required?
4. How should the S$45/t price influence—not dictate—the roadmap?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 8: Design an Employee ActNow Programme

Real-use-case lens	Singapore professional-services firm
Source lens	United Nations ActNow
Objective	Convert selected personal actions into a privacy-respecting corporate engagement programme with measurable outcomes.
Deliverable	Programme canvas, measurement plan and communication copy

Scenario

Marina Advisory has 450 employees and material Scope 3 emissions from commuting, business travel, purchased goods and food at events. Leaders want an eight-week ActNow campaign without overstating carbon savings.

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Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. Review the firm's screening footprint and identify controllable behaviour-linked sources.
2. Select three UN ActNow actions that connect to those sources.
3. Define one behaviour, one enabling policy and one barrier for each action.
4. Use the SG Carbon Calculator only for optional learning and directional reflection.
5. Define aggregate participation and operational outcome metrics.
6. Set a baseline using booking, travel, catering or waste data—not personal claims alone.
7. Design an eight-week cadence of prompt, enablement, feedback and recognition.
8. Write a privacy notice and a no-shaming participation principle.
9. Create a method to avoid double counting with formal reduction projects.
10. Draft a balanced results message showing actions, evidence and limitations.

Scenario questions

1. Which UN actions fit the firm's footprint?
2. Which outcomes can be measured credibly?
3. How will privacy and voluntary participation be protected?
4. How will you prevent double counting with travel-policy savings?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

Activity 9: Build the Board-Ready Carbon Reduction Plan

Real-use-case lens	Lion City Precision Pte Ltd
Source lens	Integrated course case using GHG Protocol, ISO/EMS and Singapore context
Objective	Integrate hotspot analysis, prioritisation, governance and action planning into a defensible 2030 plan.
Deliverable	Ten-slide board pack plus action register and management-review dashboard

Scenario

Lion City Precision manufactures components in Singapore. Its 2025 screening footprint is 10,000 tCO_2e: purchased aluminium 42%, electricity 28%, natural gas 12%, freight 9%, waste 5% and other 4%. The board is considering a 35% absolute reduction by 2030 while production grows 4% annually.

Evidence note: Company-linked facts are used only as a public case lens. Unless explicitly identified as reported, all activity figures are illustrative teaching data.

Detailed procedure

1. State the organisational and operational boundary and the 2025 baseline limitations.

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2. Calculate source contributions and identify the top hotspots.
3. Build a 2030 business-as-usual scenario with 4% annual production growth.
4. Define a precise 2030 absolute target and interim milestones.
5. Evaluate the initiative long list using carbon, cash, feasibility, strategy and confidence.
6. Select and sequence a portfolio, checking dependencies and double counting.
7. Create an initiative register with owner, budget, milestone, KPI and M&V method.
8. Design governance using executive accountability, process owners and a carbon manager.
9. Create quarterly outcome, driver, delivery and data-quality KPIs.
10. Present the plan, defend its assumptions and record peer-review changes.

Scenario questions

1. What is the business-as-usual 2030 footprint?
2. Which portfolio can credibly deliver the target?
3. What governance and data controls are essential?
4. What should management review quarterly?

Evidence and reflection

- State the boundary, period, unit, factor and source.
- Show calculations and retain intermediate values.
- Separate evidence, assumption and judgement.
- Record uncertainty and the evidence that would change the decision.
- Submit the specified deliverable and a concise management recommendation.

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Source Register

GHG Protocol Corporate Standard: https://ghgprotocol.org/corporate-standard

United Nations ActNow — Ten Actions: https://www.un.org/en/actnow/ten-actions

NEA — Singapore Carbon Tax: https://www.nea.gov.sg/our-services/climate-change-energy-efficiency/climate-change/carbon-tax

NEA — Measurement and Reporting Requirements: https://www.nea.gov.sg/our-services/climate-change-energy-efficiency/climate-change/carbon-tax/measurement-and-reporting-requirements-for-greenhouse-gas-emissions

EMA — Singapore Energy Statistics, Chapter 2: https://www.ema.gov.sg/resources/singapore-energy-statistics/chapter2

SAP Sustainability Footprint Management: https://www.sap.com/sea/products/scm/sustainability-footprint-management.html

IMD — Carbon Management: https://www.imd.org/blog/sustainability/carbon-management/

Yurtay (2025), Industry 4.0 and ERP Systems: https://www.mdpi.com/2076-3417/15/1/480

Synesgy — Carbon Footprint Guide: https://www.synesgy.com/en/esg-guide/carbon-footprint/

Green Earth — Carbon Footprints: https://www.green.earth/carbonfootprints

CarbonChain — Reducing Company Footprints: https://www.carbonchain.com/carbon-accounting/how-can-companies-reduce-carbon-footprint

SG Carbon Calculator: https://github.com/alfredang/sgcarboncalculator

Important Use Note

Regulatory rates, thresholds, factors and reporting requirements can change. Verify current primary-source requirements before making external claims, filing reports or committing capital.

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